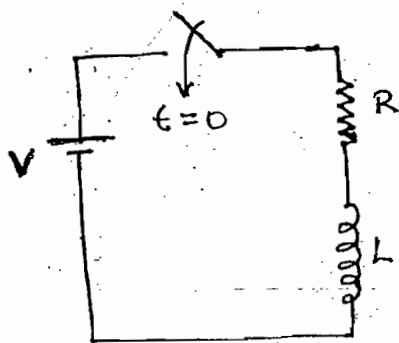


TRANSIENTS :-



$$t = 0^-, i = 0$$

$$t = 0^+, i = 0$$

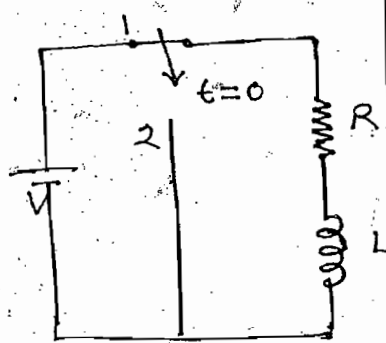
(O.C)

$$t = \infty, V_L = L \frac{di}{dt}$$

$$= 0$$

↓
S.C

$$i = \frac{V}{R}$$

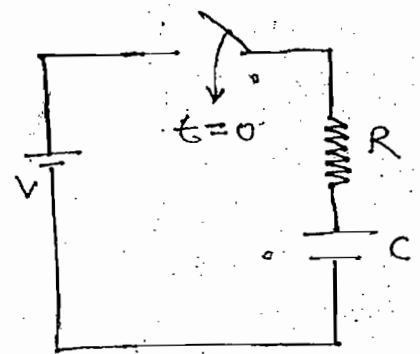


$$t = 0^-, i = I_0$$

$$t = 0^+, i = I_0$$

Inductor behaves
as current
source

$$t = \infty, i = 0$$



$$t = 0^-, V_C = 0$$

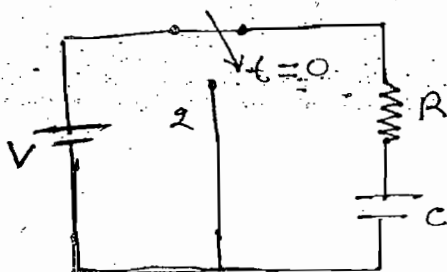
$$t = 0^+, V_C = 0$$

↓
(S.C)

$$t = \infty, V_C = V$$

$$i = 0$$

(O.C)



$$t = 0^-,$$

$$V_C = V_0 (V)$$

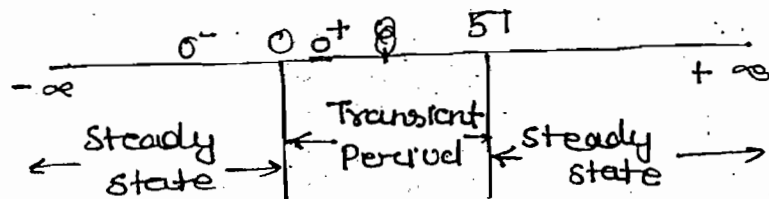
$$t = 0^+,$$

$$V_C = V_0$$

↓
Acts as voltage source

$$t = \infty$$

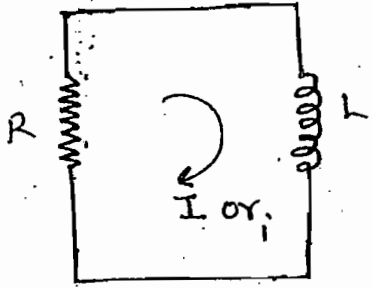
$$V_C = 0$$



- Transients are present in the circuit when circuit is subjected to any changes either by changing source magnitude or by changing circuit element provided circuit consist of any energy storage element
- When the ckt is having only resistive element then no transients are present in the N/w since
 - (i) Resistor allow sudden change of current and voltage
 - (ii) Resistor does not store any energy
- Transients are present in the N/w when the N/w is having any energy storage elements since
 - (i) Inductor does not allow sudden change of current and it stores energy in the form of magnetic field
 - (ii) Capacitor does not allow sudden change of voltage and it stores energy in the form of electric field
- Due to energy storage property inductor and capacitor are known as dynamic elements (Memory elements)
- $t = 0^-$ indicates immediately before operate the switch
- $t = 0^+$ indicates immediately after operating the switch
- $t = \infty$ indicates steady state condition after operating the switch.

Source Free RL Circuit:-

$t > 0$



$$iR + L \frac{di}{dt} = 0$$

$$iR = -L \frac{di}{dt}$$

$$\int_0^t -\frac{R}{L} dt = \int_{I_0}^{i(t)} \frac{di}{i}$$

$$\Rightarrow i(t) = I_0 e^{-\frac{Rt}{L}} \rightarrow$$

$$V_R = iR$$

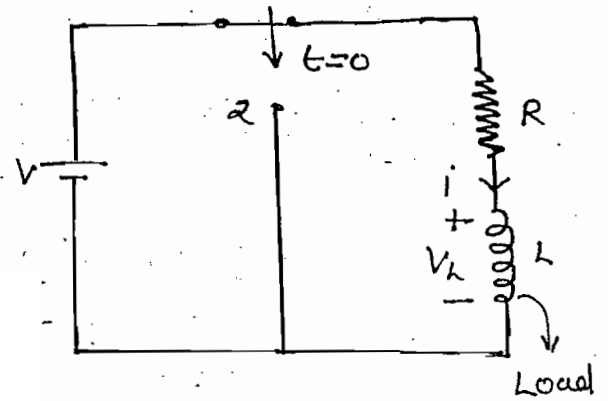
$$\Rightarrow V_R = I_0 R e^{-\frac{Rt}{L}} \rightarrow$$

$$V_L = L \frac{di}{dt}$$

$$V_L = L \frac{d}{dt} (I_0 e^{-\frac{Rt}{L}})$$

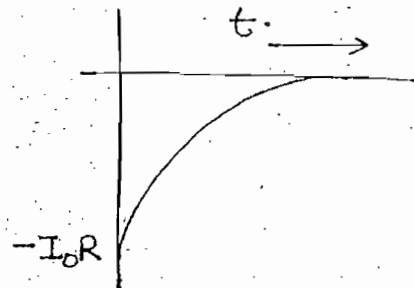
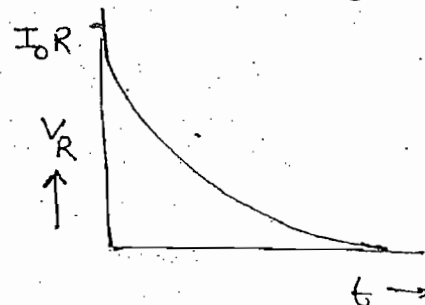
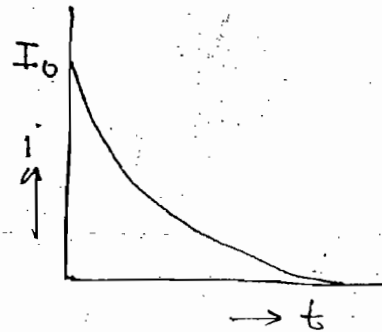
$$\Rightarrow V_L = -I_0 R e^{-\frac{Rt}{L}} \rightarrow$$

$t \rightarrow 0$



$$t = 0^-, i = I_0$$

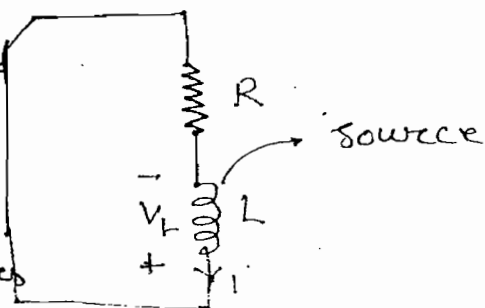
$$t = 0^+, i = I_0$$



$t > 0$

Note:-

When switch is transferred from position 1 to position 2 current direction of the inductor remains same but voltage across inductor polarities are reversed.



RL Circuit with Source:-

By KVL

$$V = iR + L \frac{di}{dt}$$

Divide whole equation by L

$$\frac{V}{L} = \frac{iR}{L} + \frac{di}{dt}$$

$$\Rightarrow \boxed{\frac{di}{dt} + \frac{R}{L} i = \frac{V}{L}}$$

$$i(t) = C.F + P.I$$

C.F $\rightarrow$ Transient response or source free response or Natural Response

$$\frac{di}{dt} + \frac{R}{L} i = 0 \Rightarrow i(t) = A e^{-\frac{Rt}{L}}$$

P.I $\rightarrow$ Steady state response or final value or forced Response

$$i(t) = C.F + P.I$$

$$i(t) = A e^{-\frac{Rt}{L}} + \frac{V}{R}$$

$$t = 0^- \quad , \quad i = 0$$

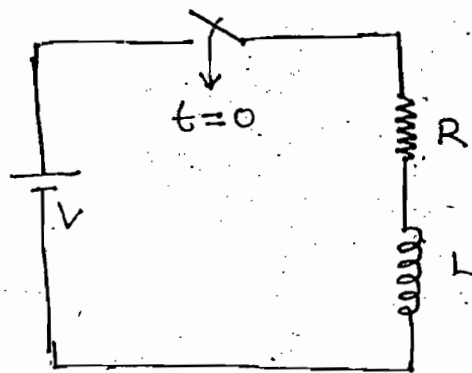
$$t = 0^+ \quad , \quad i = 0$$

$$\Rightarrow 0 = A + \frac{V}{R}$$

$$\Rightarrow A = 0 - \frac{V}{R}$$

$$\Rightarrow \boxed{A = i(0^+) - i(\infty)}$$

$\rightarrow$ Applicable for RL or RC circuit with source.



$$i(t) = C.F + P.I$$

$$i(t) = -\frac{V}{R} e^{-Rt/L} + \frac{V}{R}$$

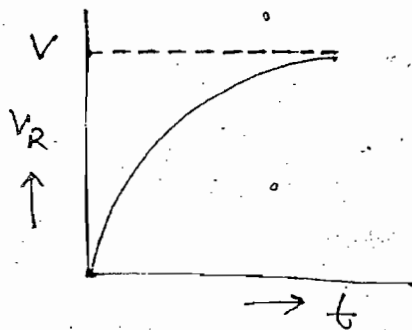
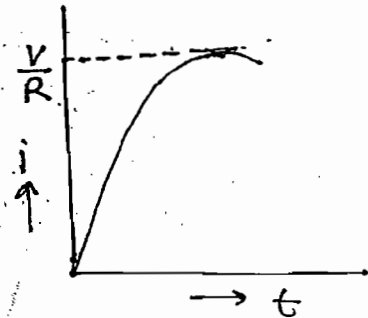
$$\Rightarrow i(t) = [i(0^+) - i(\infty)] e^{-Rt/L} + i(\infty)$$

→ Valid for RL and RC circuit

$$i(t) = \frac{V}{R} (1 - e^{-Rt/L}) \rightarrow$$

$$V_R = iR$$

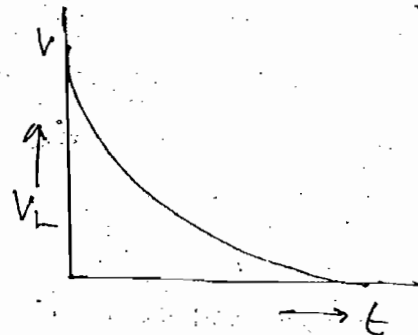
$$\Rightarrow V_R = V(1 - e^{-Rt/L})$$



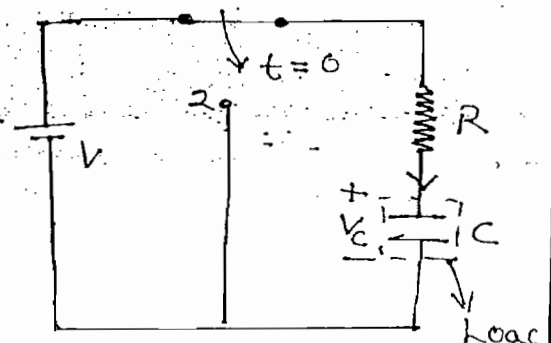
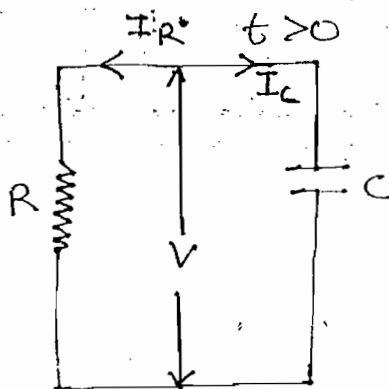
$$V_L = L \frac{di}{dt}$$

$$V_L = L \frac{d}{dt} \left[\frac{V}{R} (1 - e^{-Rt/L}) \right]$$

$$V_L = V_0 e^{-Rt/L}$$



Source Free RC Circuit:-



$$t = 0^-, V_C = V_0$$

$$t = 0^+, V_C = V_0$$

V = Virtual voltage source

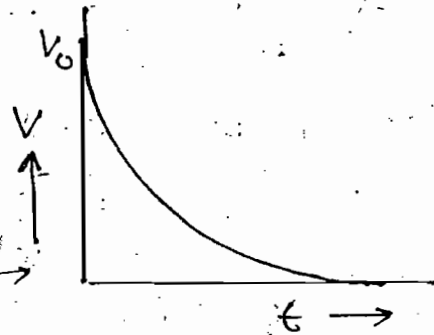
$$I_R + I_C = 0$$

$$\Rightarrow \frac{V}{R} + C \frac{dV}{dt} = 0$$

$$\Rightarrow \frac{V}{R} = -C \frac{dV}{dt}$$

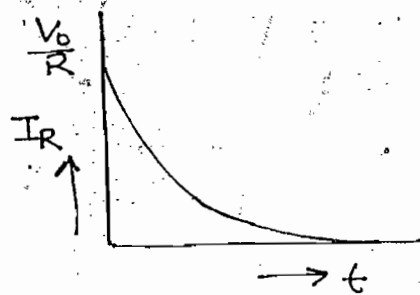
$$\Rightarrow \int_0^t -\frac{1}{RC} dt = \int_{V_0}^{V(t)} \frac{dV}{V}$$

$$\Rightarrow V(t) = V_0 e^{-t/RC}$$



$$I_R = \frac{V}{R}$$

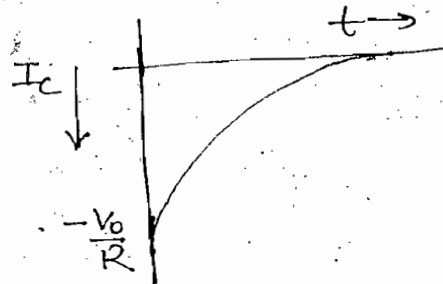
$$\Rightarrow I_R = \frac{V_0}{R} e^{-t/RC}$$



$$I_C = C \frac{dV}{dt}$$

$$I_C = C \frac{d}{dt} (V_0 e^{-t/RC})$$

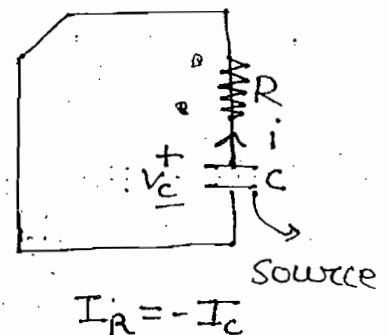
$$\Rightarrow I_C = -\frac{V_0}{R} e^{-t/RC}$$



$t > 0$

Note:-

When switch is transferred from position 1 to position 2 voltage across the capacitor will remain same but current direction of the capacitor is reversed.



RC Circuit with Source

By KVL

$$V = iR + \frac{1}{C} \int i dt$$

Diff. w.r.t t

$$0 = R \frac{di}{dt} + \frac{i}{C}$$

Divide both sides by R we get

$$\frac{di}{dt} + \frac{i}{RC} = 0$$

$$i(t) = CF + PI$$

CF $\rightarrow$ Transient Response

$$\frac{di}{dt} + \frac{i}{RC} = 0$$

$$i(t) = A e^{-t/RC}$$

PI $\rightarrow$ steady state response

$$C \rightarrow \infty$$

$$i = 0$$

$$i(t) = CF + PI$$

$$i(t) = A e^{-t/RC} + 0$$

$$A = i(0^+) - i(\infty)$$

$$t = 0^-, V_C = 0$$

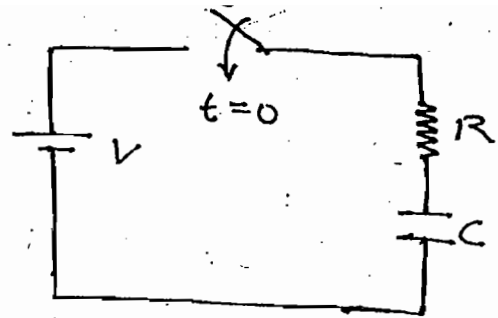
$$t = 0^+, V_C = 0$$

By KVL

$$t = 0^+ \quad V = V_R + V_C \\ = iR + 0$$

$$\text{at } t \rightarrow \infty \quad i = \frac{V}{R}$$

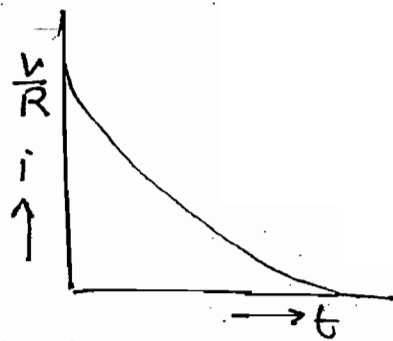
$$i(t) = CF + PI$$



$$A = i(0^+) - i(\infty)$$

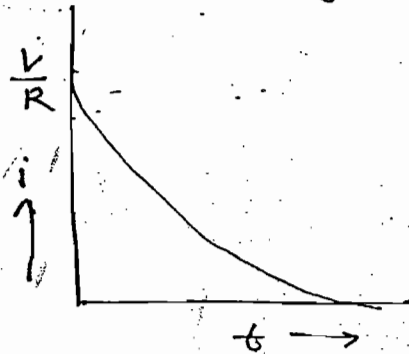
$$A = \frac{V}{R} - 0$$

$$i(t) = \frac{V}{R} e^{-t/RC}$$



$$V_R = iR$$

$$V_R = V e^{-t/RC}$$



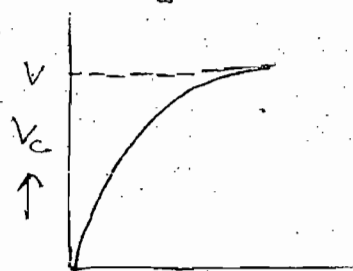
$$V_C = \frac{1}{C} \int_0^t i dt$$

$$V_C = \frac{1}{C} \int_0^t \frac{V}{R} e^{-t/RC} dt$$

$$V_C = -V e^{-t/RC} + V$$

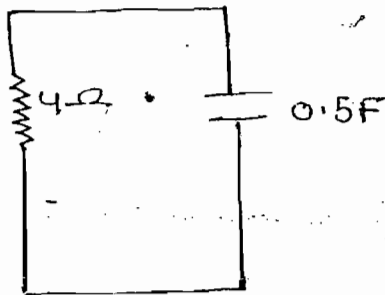
$$V_C(t) = [V_C(0^+) - V_C(\infty)] e^{-t/RC} + V_C(\infty)$$

$$V_C(t) = V(1 - e^{-t/RC})$$



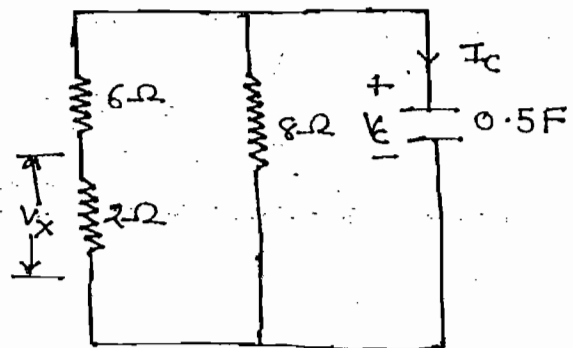
Ques:- Find response of V_C , I_C and V_x when initial voltage of the capacitor is 3V

Soln:-



$$V_C = V_0 e^{-t/RC}$$

$$V_C = 3 e^{-t/2}$$



$$V_x = V_c \frac{2}{2+t} \Rightarrow V_x = 3e^{-t/2} \cdot \frac{1}{4}$$

$$\Rightarrow V_x = \frac{3}{4} e^{-t/2}$$

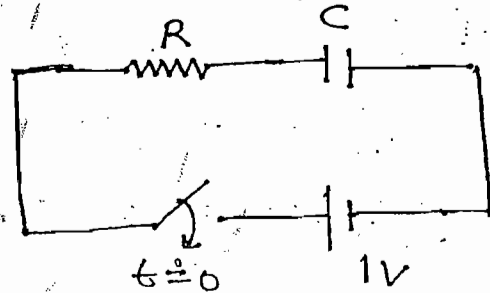
$$I_c = C \frac{dV_c}{dt}$$

$$I_c = \frac{1}{2} \frac{d(3e^{-t/2})}{dt} = 3 \left(\frac{1}{2}\right) \left(-\frac{1}{2}\right) e^{-t/2}$$

$$\Rightarrow \boxed{I_c = -\frac{3}{4} e^{-t/2}} \quad \text{Ans.}$$

Ques:- Find rate of rise of voltage across the capacitor at $t = 0^+$

(a) RC (b) $\frac{1}{RC}$ (c) 0 (d) $2RC$



Soln:-

$$V_c = -V e^{-t/RC} + V$$

$$\frac{dV_c}{dt} = (-V) \left(-\frac{1}{RC}\right) e^{-t/RC} + 0$$

$$\frac{dV_c}{dt} = \frac{V}{RC} e^{-t/RC}$$

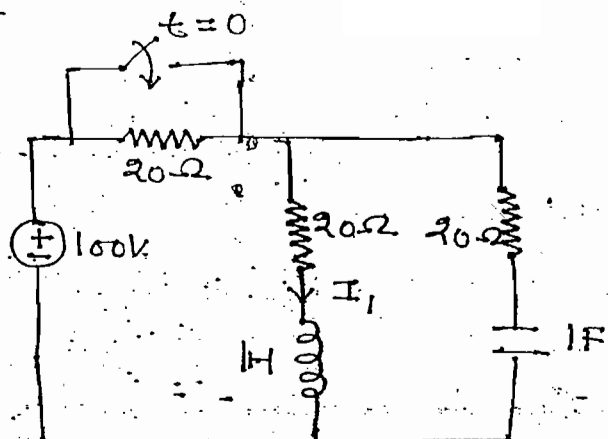
$$t = 0^+$$

$$\frac{dV_c}{dt} (0^+) = \frac{V}{RC} = \frac{1}{RC}$$

Ques:- Find $\frac{dI_1}{dt}$ at $t = 0^+$

Soln:- Step-(1): -

Develop eq. circuit at $t = 0^-$ and find initial current of conductor and initial voltage of capacitor



→ In the above ckt at $t = 0^-$ for DC source inductor behaves as S.C. and C behaves as O.C

$$i_L(0^-) = \frac{100}{20+20} = 2.5A$$

$$V_C(0^-) = 2.5 \times 20 = 50$$

Step-2:-

Develop eq. ckt at $t=0^+$ and indicate initial current of the inductor and initial voltage of the capacitor.

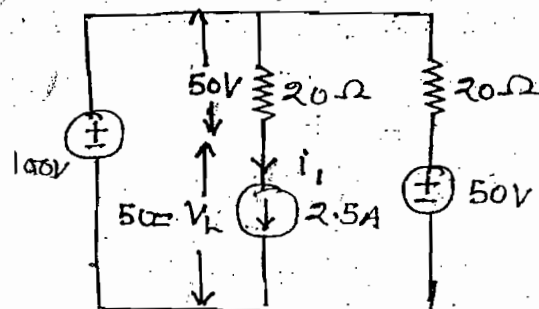
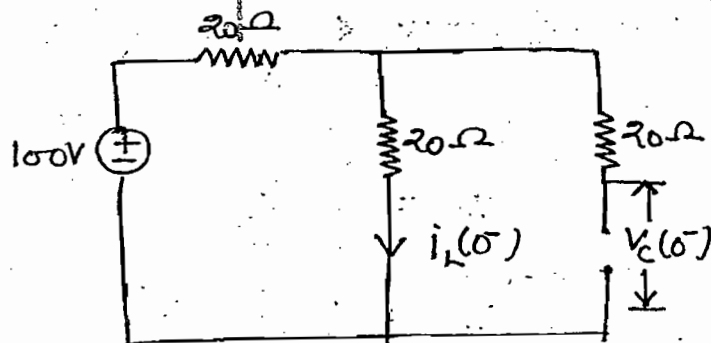
$t=0^+$

$$i_L(0^+) = i_L(0^-) = 2.5A$$

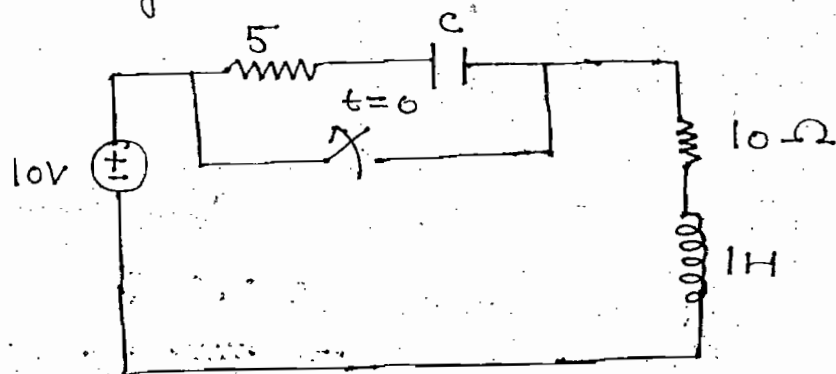
$$V_C(0^+) = V_C(0^-) = 50V$$

$$V_L = L \frac{di}{dt}$$

$$\Rightarrow 50 = 1 \frac{di}{dt} \Rightarrow \frac{di}{dt} = 50 A/s \quad Ans$$



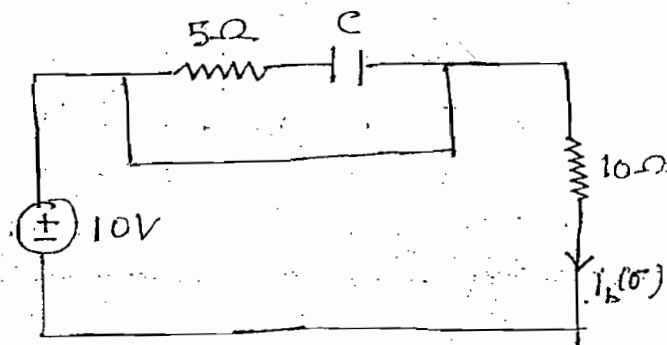
ques:- Find voltage across the switch at $t=0^+$



Soln:- At $t=0^-$

→ Develop equivalent circuit at $t=0^-$

→ In the above ckt. at $t=0^-$ capacitor is uncharged & for DC source inductor behaves as a S.C



$$i_L(0^-) = \frac{10}{10} = 1A, \quad V_C(0^-) = 0$$

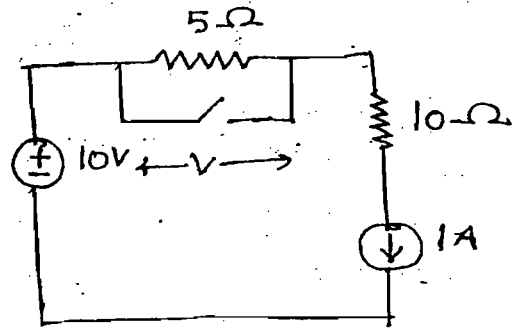
Step-2:-

At $t = 0^+$

$$i_L(0^+) = i_L(0^-) = 1A$$

$$V_C(0^+) = V_C(0^-) = 0$$

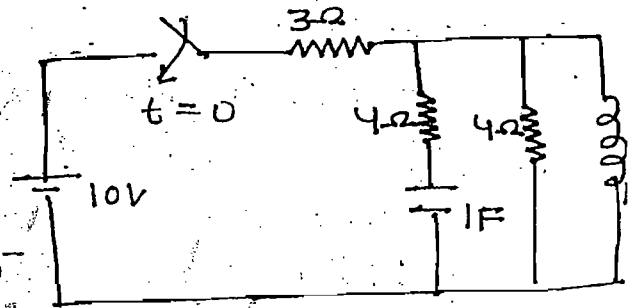
$$V(0^+) = 5 \times 1 = 5V$$



ques:- Find I in capacitor and voltage across inductor at $t = 0^+$

Soln:- Step (I):-

Develop eq. ckt. at $t = 0^-$



→ In the above ckt at $t = 0^-$ capacitor and inductor are uncharged elements.

$$t = 0^-, \quad V_C(0^-) = 0$$

$$i_L(0^-) = 0$$

Step-(II):-

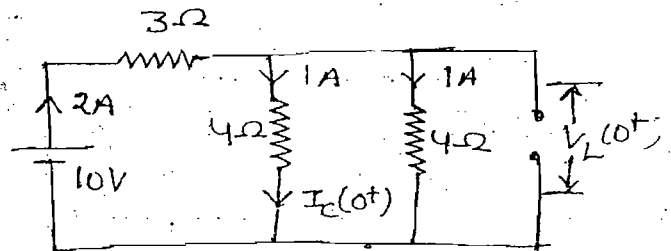
At $t = 0^+$

$$i_L(0^+) = i_L(0^-) = 0$$

$$V_C(0^+) = V_C(0^-) = 0$$

$$I_C(0^+) = 1A$$

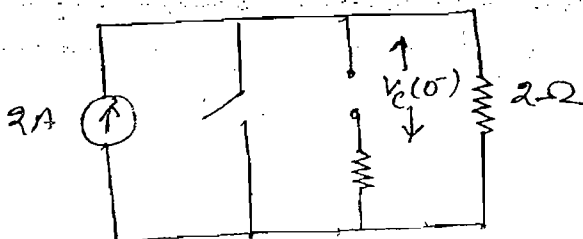
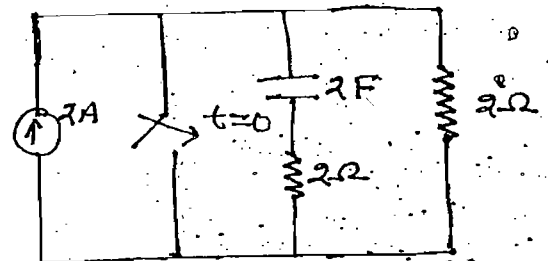
$$V_L(0^+) = 4 \times 1 = 4V$$



ques:- Find initial voltage and final voltage of the capacitor

Soln:- Step (I):-

$$t = 0^-$$



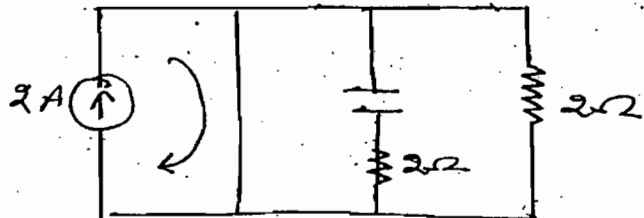
$$V_C(0^-) = 2 \times 2 = 4V$$

Step-(ii):-

$$\text{At } t = 0^+$$

$$V_C(0^+) = V_C(0^-) = 4V$$

$$V_C(\infty) = 0$$

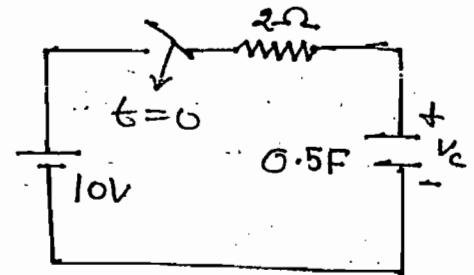


Ques:- Find

(i) Energy of Capacitor at $t = \infty$

(ii) Current in the capacitor at $t = 1\text{sec}$

(iii) Energy of the resistor from 0 to ∞ interval

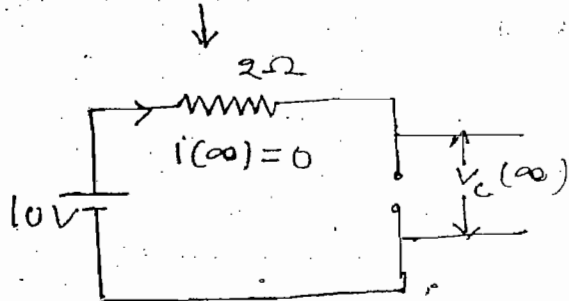


When initial charge of the capacitor is $10C$

Soln:- At $t = 0^+$ ($\because Q_{in} = 10C$)

$$V_0 = \frac{Q_0}{C} = \frac{10}{0.5} \Rightarrow V_0 = 20$$

At $t = \infty$



$$V_C(\infty) = 10V$$

$$W_C(\infty) = \frac{1}{2} C V_C^2(\infty) = \frac{1}{2} \cdot \frac{1}{2} \cdot (10)^2 = 25J$$

$$(ii) i(t) = [i(0^+) - i(\infty)] e^{-t/RC} + i(\infty)$$

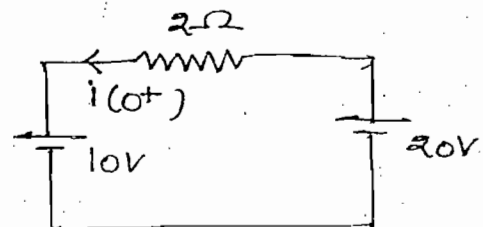
$$\Rightarrow i(t) = [-5 - 0] e^{-t/1} + 0$$

$$i(t) = -5 e^{-t}$$

At $t = 1\text{sec}$

$$i(1s) = -5 e^{-1} = -\frac{5}{e}$$

$$= -1.84$$



(iii)

$$W_R = \int_0^{\infty} P dt = \int_0^{\infty} i^2 R dt = \int_0^{\infty} (-5e^{-t})^2 2 dt$$

$$\Rightarrow W_R = 25J \quad , \text{Ans.}$$

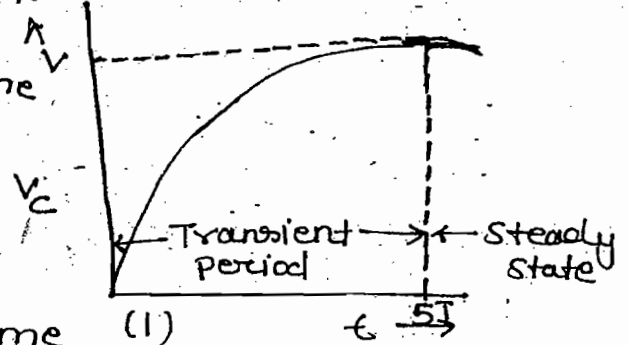
Time Constant :-

→ Time constant can be define either w.r.t charging or discharging action of energy storage element

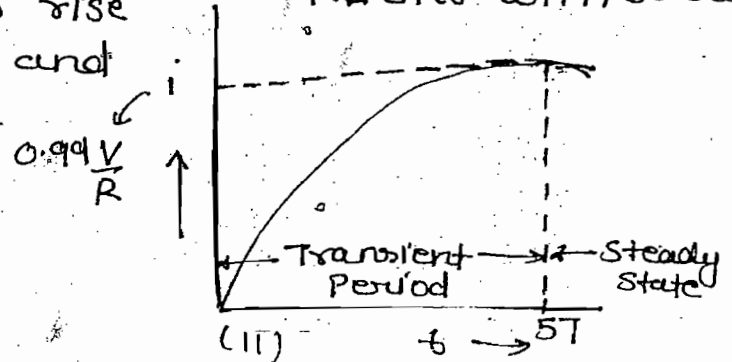
→ Time constant is the time taken for response to rise 63.2% of max. value and it is given by

$$\boxed{\begin{array}{l} T = \frac{L}{R} \text{ sec.} \\ T = RC \end{array}}$$

RC ckt with source



RL ckt with source



$$V_C(t) = V(1 - e^{-t/T})$$

$$(I) V_C(t) = V(1 - e^{-t/RC})$$

$$(I) t = T \quad V_C = V(1 - e^{-1}) = 0.632V$$

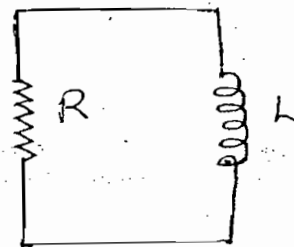
$$(II) i(t) = \frac{V}{R}(1 - e^{-Rt/L})$$

$$t = 5T \quad V_C = V(1 - e^{-5}) = 0.99V$$

$$R_1 = 10\Omega \quad P_1 = I^2 R_1 \quad t = 10\text{sec}$$

$$R_2 = 20\Omega \quad P_2 = i^2 R_2 \quad t = 7\text{sec}$$

$$R_3 = 100\Omega \quad P_3 = i^2 R_3 \quad t = 2\text{sec}$$



$$\text{Hence } T = \frac{L}{R}$$

$$R_1 = 10\Omega$$

$$P_1 = \frac{V^2}{R_1}$$

$$t = 10\text{sec}$$

$$R_2 = 20\Omega$$

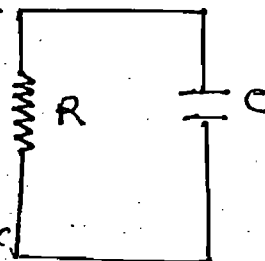
$$P_2 = \frac{V^2}{R_2}$$

$$t = 15\text{sec}$$

$$R_3 = 100\Omega$$

$$P_3 = \frac{V^2}{R_3}$$

$$t = 50\text{sec}$$

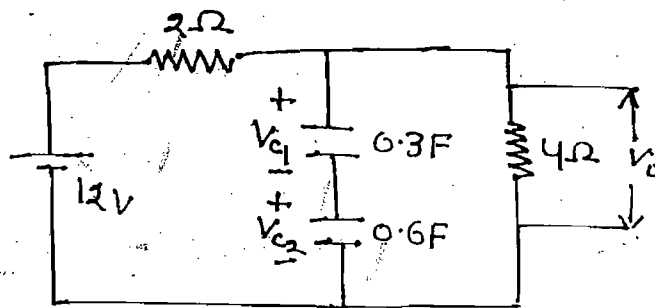


$$\text{Hence } T = RC$$

Ques:- Find V_o response for $t > 0$

$$V_{C_1}(0) = 24\text{V}$$

$$V_{C_2}(0) = 6\text{V}$$



$$(a) 8 + 22e^{-3.75t}$$

$$(b) 8 + 22e^{-t/3.75}$$

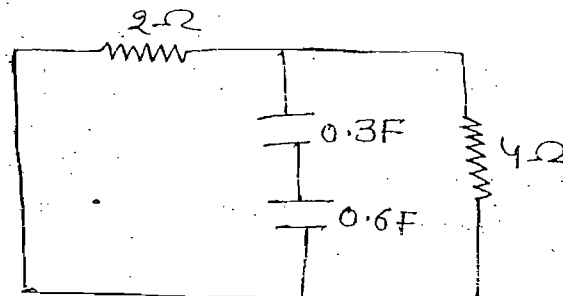
$$(c) 8 + 22e^{-t/1.2}$$

$$(d) 8 + 22e^{-1.2t}$$

Soln:- For time constant, deactivate all the independent sources

$$R_{eq} = \frac{2 \times 4}{2 + 4}$$

$$C_{eq} = \frac{0.3 \times 0.6}{0.3 + 0.6}$$



$$T^0 = R_{eq} C_{eq} = RC$$

$$V(t) = [V(0^+) - V(\infty)] e^{-t/RC} + V(\infty)$$

$$V_o(t) = [V_o(0^+) - V_o(\infty)] e^{-t/RC} + V_o(\infty)$$

$$V_o = V_{C_1} + V_{C_2}$$

$$t = 0^+$$

$$V_o(0^+) = V_{C_1}(0^+) + V_{C_2}(0^+) = 24 + 6 = 30$$

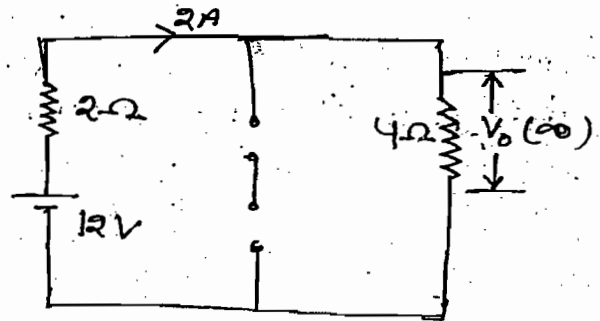
$$\text{At } t = \infty$$

$$\text{At } t = \infty$$

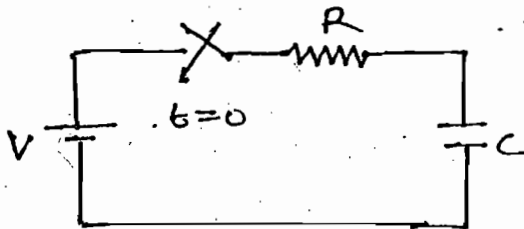
$$V_0(\infty) = 2 \times 4 = 8$$

$$V_0(t) = [30 - 8] e^{-3.75t} + 8$$

$$= 8 + 22 e^{-3.75t}$$



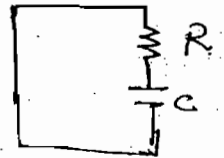
(I)



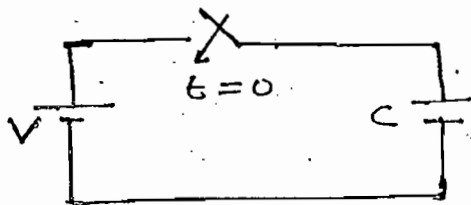
$$\text{At } t=0^-, V_C = 0$$

$$t=0^+, V_C = 0$$

$$T = RC$$



(II)



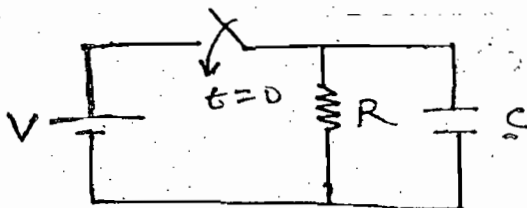
$$t=0^-, V_C = 0$$

$$t=0^+, V_C = V$$

$$T = RC = 0$$



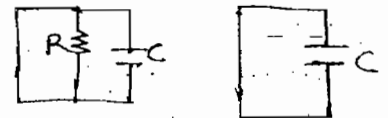
(III)



$$t=0^-, V_C = 0$$

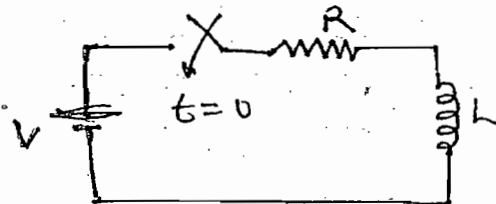
$$t=0^+, V_C = V$$

$$T = RC = 0$$



$$T = RC = 0$$

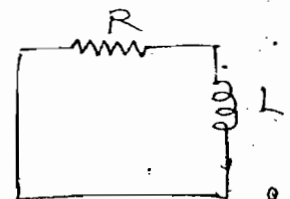
(IV)



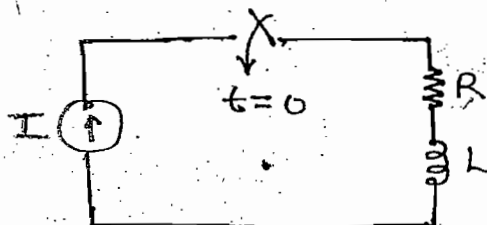
$$t=0^-, i = 0$$

$$t=0^+, i = 0$$

$$T = L/R$$



(V)

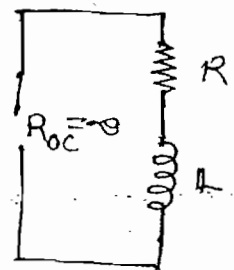


$$t=0^-, i = 0$$

$$t=0^+, i = I$$

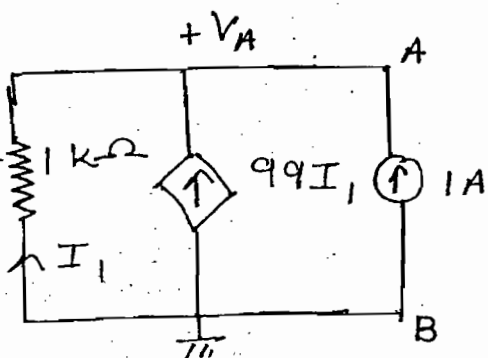
$$R_{eq} = R_{oc} + R = \infty$$

$$T = \frac{L}{R_{eq}} = 0$$



Ques:- Find time constant of the circuit shown

Soln:-

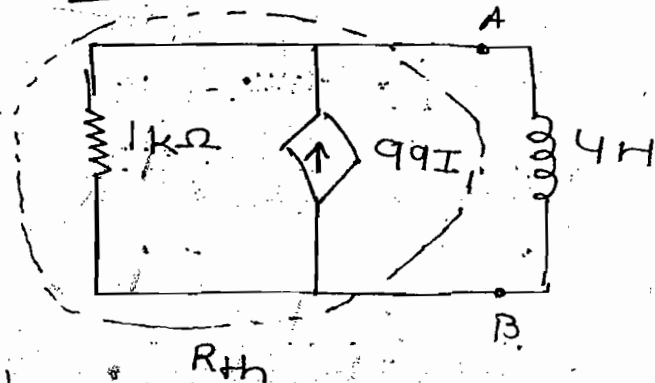
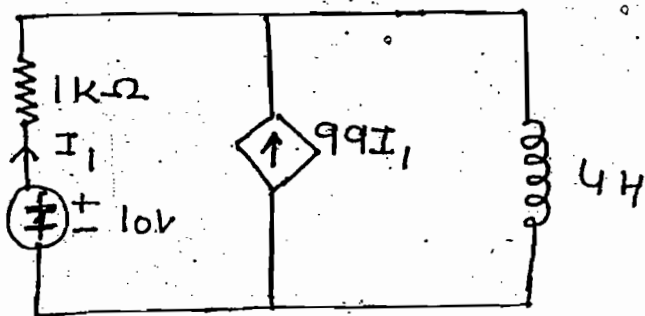


$$\frac{V_A}{1 \times 10^3} = 99I_1 + 1 \quad \text{--- (i)}$$

$$I_1 = -\frac{V_A}{1 \times 10^3} \quad \text{--- (ii)}$$

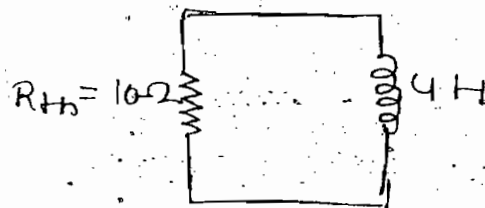
$$R_{th} = \frac{V_A}{I_S} = \frac{10}{1} = 10 \Omega$$

$$T = \frac{L}{R_{th}} = \frac{4}{10}$$



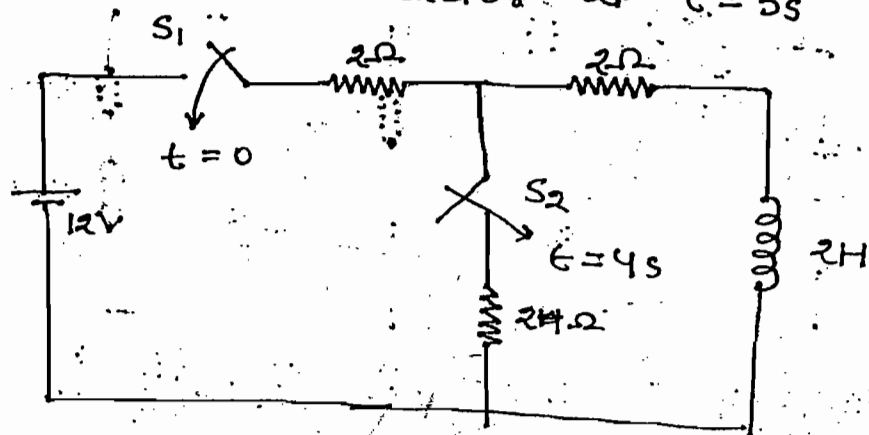
From (i) & (ii)

$$V_A = 10V$$



Lecture -10

Ques:- Find current in the inductor at $t = 5s$



Soln:-

S_1

$$i(t) = \frac{V}{R} (1 - e^{-Rt/L})$$

$$i(t) = \frac{12}{4} (1 - e^{-2t}) \Rightarrow i(t) = 3(1 - e^{-2t})$$

$t > 4s$, S_1 and $S_2 \rightarrow$ operated

$$i(t) = [i(0^+) - i(\infty)]e^{-Rt/L} + i(\infty)$$

$$\Rightarrow i(t) = [i(4) - i(\infty)]e^{-R'(t-4)} + i(\infty) \quad (1)$$

$t = 4$

$$i(4) = 3[1 - e^{-2(4)}] \approx 2.99A$$

$$t = \infty$$

$$i(\infty) = 2A$$

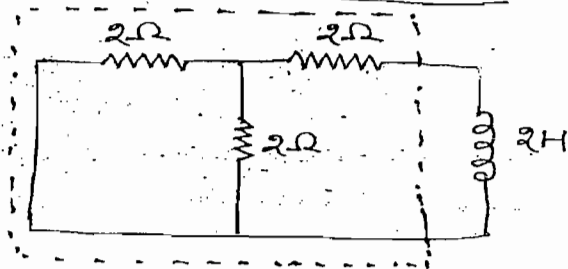
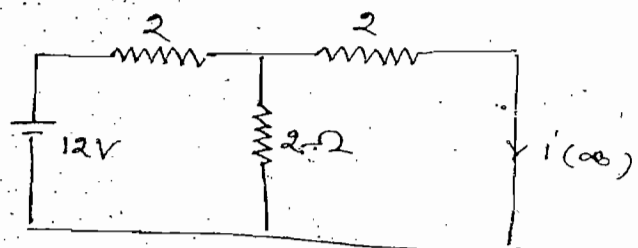
Time constant for $T > 4s$

$$R' = 3 \quad (11)$$

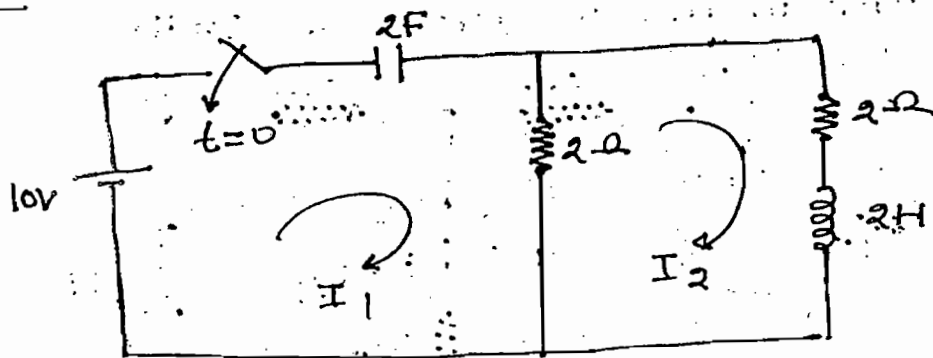
From - (1) & (11)

$$i(t) = [2.99 - 2]e^{-3/2(t-4)} + 2$$

$$t = 5 \text{ sec.}$$



Prob:-



Find $i_1(0^+)$, $i_2(0^+)$, $V_C(0^+)$, $\frac{di_1(0^+)}{dt}$, $\frac{di_2(0^+)}{dt}$, $\frac{d^2 i_1(0^+)}{dt^2}$,

$$\frac{d^2 i_2(0^+)}{dt^2}$$

Soln:-

At $t=0^-$

$$i_1(0^-) = 0 \quad i_2(0^-) = 0 \quad V_C(0^-) = 0$$

At $t=0^+$

$$i_2(0^+) = 0 \quad V_C(0^+) = 0 \quad i_1(0^+) = \frac{10}{2} = 5A$$

$$(1) \quad 4I_2 + 2 \frac{di_2}{dt} - 2I_1 = 0 \quad \rightarrow t=0^+$$

$$4(0) + 2 \frac{di_2}{dt} - 2(5) = 0$$

$$\Rightarrow \frac{di_2}{dt} = 5 A/s$$

$$-10 + \frac{1}{2} \int i_1 dt + 2(I_1 - I_2) = 0$$

Diff. w.r.t t

$$0 + \frac{i_1}{2} + 2 \left[\frac{di_1}{dt} - \frac{di_2}{dt} \right] = 0 \quad \text{--- (II)}$$

$$\Rightarrow \frac{5}{2} + 2 \left[\frac{di_1}{dt} - 5 \right] = 0$$

$$\Rightarrow \frac{5}{2} - 10 + 2 \frac{di_1}{dt} = 0$$

$$\Rightarrow \frac{di_1}{dt} = \frac{15}{4} = 3.75 A/s$$

Diff eq-(i) w.r.t t

$$4 \frac{di_2}{dt} + 2 \frac{d^2 i_2}{dt^2} - 2 \frac{di_1}{dt} = 0$$

$$\Rightarrow \frac{d^2 i_2}{dt^2} = -6.25 \text{ A/s}^2$$

Diff. eq-(ii), w.r.t t

$$\frac{1}{2} \frac{di_1}{dt} + 2 \left[\frac{d^2 i_1}{dt^2} - \frac{d^2 i_2}{dt^2} \right] = 0$$

$$\Rightarrow \frac{d^2 i_1}{dt^2} = -7.18 \text{ A/s}^2$$

RLC Series Circuit with AC Excitation :-

By KVL

$$V = IR + L \frac{di}{dt} + \frac{1}{C} \int i dt$$

Diff w.r.t t

$$0 = R \frac{di}{dt} + L \frac{d^2 i}{dt^2} + \frac{i}{C}$$

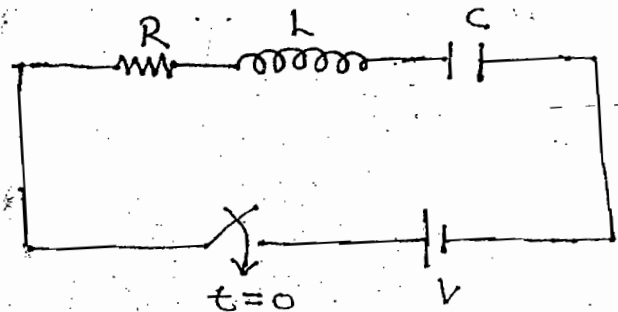
Divide both sides by L we get

$$\boxed{\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = 0}$$

$$\left(s^2 + \frac{R}{L} s + \frac{1}{LC} \right) i = 0 \quad \left(\because \frac{d}{dt} = s \right)$$

$$s_1, s_2 = \frac{-\frac{R}{L} \pm \sqrt{\left(\frac{R}{L}\right)^2 - \frac{4}{LC}}}{2}$$

$$\boxed{s_1, s_2 = \frac{-R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$



Case - (I) :-

$$\left(\frac{R}{2L}\right)^2 > \frac{1}{LC}$$

overdamping

$$\alpha = -\frac{R}{2L} \quad \beta = \sqrt{\left(\frac{R}{2L}\right)^2 - \left(\frac{1}{LC}\right)}$$

$$i(t) = C_1 e^{(\alpha-\beta)t} + C_2 e^{(\alpha+\beta)t}$$

Case - (II) :-

$$\left(\frac{R}{2L}\right)^2 = \frac{1}{LC} \rightarrow \text{Critical damping}$$

$$i(t) = (C_1 + C_2 t) e^{\alpha t}$$

Case - (III) :-

$$\left(\frac{R}{2L}\right)^2 < \frac{1}{LC} \rightarrow \text{Under damping}$$

$$\alpha = -\frac{R}{2L}$$

$$\beta = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

$$i(t) = (C_1 \cos \beta t + C_2 \sin \beta t) e^{\alpha t}$$

$$\text{Damping Coefficient} = \frac{R}{2L}$$

$$\text{Time Constant} = \frac{1}{\text{Damping Coefficient}} = \frac{2L}{R}$$

Case - (IV) :-

$$R = 0 \rightarrow \text{Underdamping}$$

$$\alpha = 0, \quad \beta = \frac{1}{\sqrt{LC}}$$

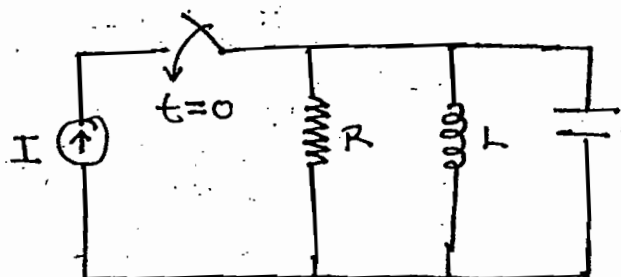
$$i(t) = C_1 \cos \beta t + C_2 \sin \beta t$$

RLC Parallel Circuit with AC Excitation:-

$$I = \frac{V}{R} + C \frac{dV}{dt} + \frac{1}{L} \int V dt$$

Diff. w.r.t. t

$$0 = \frac{1}{R} \frac{dV}{dt} + C \frac{d^2 V}{dt^2} + \frac{V}{L}$$



Divide the whole equation by C , we get

$$\frac{d^2 V}{dt^2} + \frac{1}{RC} \frac{dV}{dt} + \frac{V}{LC} = 0$$

$$\left(D^2 + \frac{1}{RC} D + \frac{1}{LC} \right) V = 0, \quad \left(\because D = \frac{d}{dt} \right)$$

$$D_1, D_2 = \frac{-\frac{1}{RC} \pm \sqrt{\left(\frac{1}{RC}\right)^2 - \frac{4}{LC}}}{2}$$

$$D_1, D_2 = \frac{-1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$$

Case - (I) :-

$$\left(\frac{1}{2RC} \right)^2 > \frac{1}{LC} \rightarrow \text{over damping}$$

$$\alpha = -\frac{1}{2RC}$$

$$\beta = \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$$

$$V(t) = C_1 e^{(\alpha - \beta)t} + C_2 e^{(\alpha + \beta)t}$$

Case - (II) :-

$$\left(\frac{1}{2RC} \right)^2 = \frac{1}{LC} \rightarrow \text{Critical damping}$$

$$V(t) = (C_1 + C_2 t) e^{\alpha t}$$

Case-(III) :-

$$\left(\frac{1}{2RC}\right)^2 < \frac{1}{LC} \rightarrow \text{Under damping}$$

$$\alpha = -\frac{1}{2RC}$$

$$\beta = \sqrt{\frac{1}{LC} - \left(\frac{1}{2RC}\right)^2}$$

$$V(t) = (C_1 \cos \beta t + C_2 \sin \beta t) e^{\alpha t}$$

$$\text{Damping coefficient} = \frac{1}{2RC}$$

$$\text{Time constant} = \frac{1}{\text{Damping coefficient}} = 2RC$$

Case-(IV) :-

$$G = 0$$

$$\frac{1}{R} = 0 \Rightarrow R = \infty$$

$\rightarrow$ Underdamping

$$\alpha = 0, \beta = \frac{1}{\sqrt{LC}}$$

$$V(t) = C_1 \cos \beta t + C_2 \sin \beta t$$

Note:-

1. Steady state current response of RLC series circuit with AC excitation = 0

2. At $t = \infty$,

$L \rightarrow$ Uncharged ($I = 0$)

$C \rightarrow$ Charged ($V_C = V$)

3.

$$\beta = \frac{R}{2} \sqrt{\frac{C}{L}}$$

$\rightarrow$ Series

$\beta > 1 \rightarrow$ overdamping

$\beta < 1 \rightarrow$ Underdamping

$\beta = 1 \rightarrow$ Critical damping

$\beta = 0 \rightarrow$ Undamping

$$\zeta = \frac{1}{2R} \sqrt{\frac{L}{C}}$$

→ Parallel

4. For over and critical damping system no oscillations are present

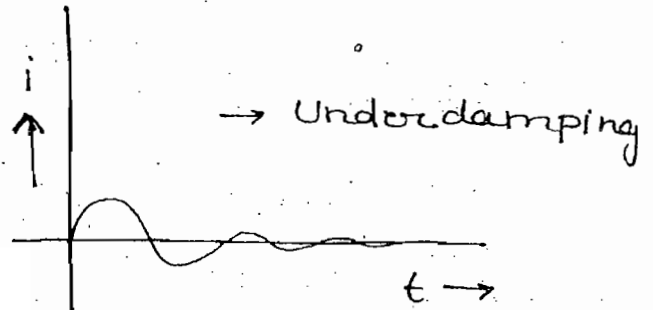
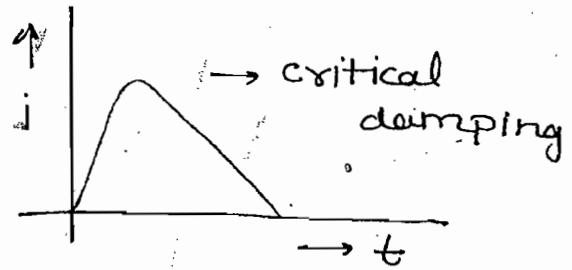
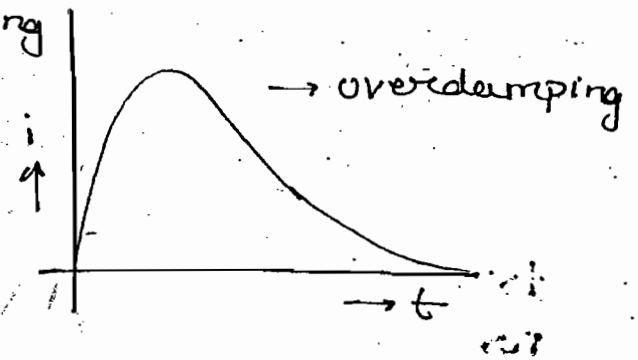
$$\frac{1}{2R} \sqrt{\frac{L}{C}} \geq 1 \rightarrow \text{Parallel}$$

$$\frac{R}{2} \sqrt{\frac{C}{L}} \geq 1 \rightarrow \text{Series}$$

$$\zeta \geq 1$$

→ In the underdamping system more than 1 oscillation are present

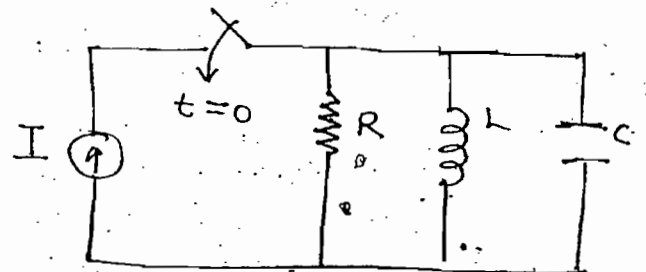
$$\zeta < 1$$



5. At $t = \infty$

L → Charged ($I_L = I$)

C → Uncharged ($V_C = 0$)

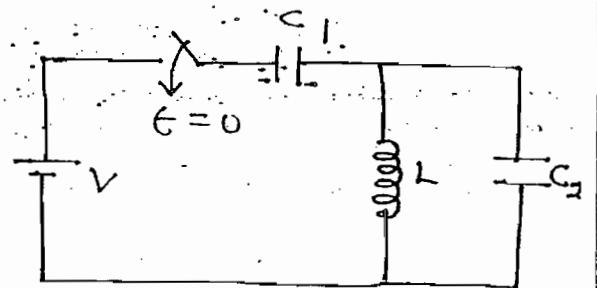


6. $t = \infty$

C_1 → Charged ($V_{C_1} = V$)

C_2 → Uncharged ($V_{C_2} = 0$)

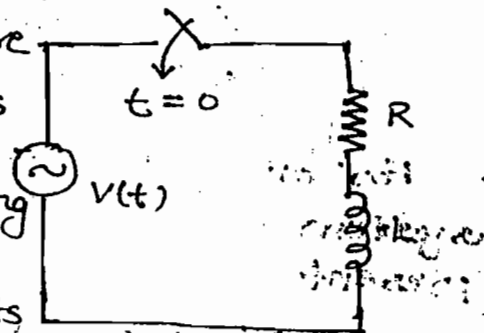
L → Uncharged ($I_L = 0$)



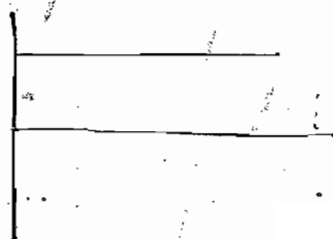
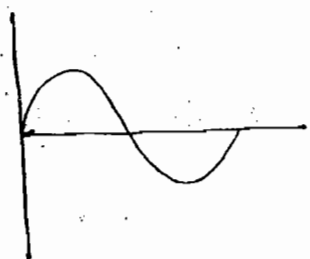
A.C Transients :-

→ DC transients intensity is more than intensity of AC transients

→ Based on selection of operating frequency and switching operation and circuit elements it is possible to obtain transient free response. But in AC circuit it is not possible to obtain transient free response



$$V(t) = V_m \sin(\omega t + \theta)$$



By KVL

$$V = iR + L \frac{di}{dt}$$

Divide both sides by L , we get

$$\boxed{\frac{di}{dt} + \frac{R}{L}i = \frac{V}{L}}$$

$$i(t) = C.F + P.I$$



Transient Response

C.F :-

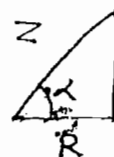
$$\frac{di}{dt} + \frac{R}{L}i = 0$$

$$\Rightarrow i(t) = A e^{-\frac{R}{L}t}$$

P.I → Steady state response

$$i = \frac{V}{Z \angle \alpha} = \frac{V L}{Z}$$

$$\boxed{i(t) = \frac{V_m}{Z} \sin(\omega t + \theta - \alpha)}$$



$$X_L = \omega L$$

$$\alpha = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

$$i(t) = C.F + P.I$$

$$i(t) = A e^{-\frac{R}{L}t} + \frac{V_m}{Z} \sin(\omega t + \theta - \alpha)$$

$$t = 0^- , \quad i = 0$$

$$t = 0^+ , \quad i = 0$$

$$0 = A + \frac{V_m}{Z} \sin(\theta - \alpha)$$

$$\Rightarrow \boxed{A = -\frac{V_m}{Z} \sin(\theta - \alpha)}$$

$$i(t) = \underbrace{-\frac{V_m}{Z} \sin(\theta - \alpha) e^{-\frac{R}{L}t}}_{T.R.} + \underbrace{\frac{V_m}{Z} \sin(\omega t + \theta - \alpha)}_{S.R.}$$

Case-(I) :-

$$V(t) = V_m \sin(\omega t + \theta) , \quad t = 0$$

$$\theta - \alpha = 0$$

$$\theta = \alpha \Rightarrow$$

$$\boxed{\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)}$$

→ Condition for transient free response

Case-(II) :-

$$V(t) = V_m \cos(\omega t + \theta) , \quad t = 0$$

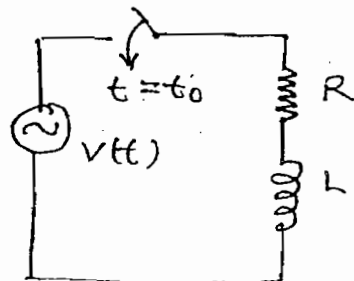
$$\boxed{\theta = \tan^{-1}\left(\frac{\omega L}{R}\right) + \frac{\pi}{2}}$$

→ condition for transient free response

Case-(III) :-

$$t = t_0^- \quad i = 0$$

$$t = t_0 \quad i = 0$$



$$0 = A + \frac{V_m}{Z} \sin(\omega t_0 + \theta - \alpha)$$

$$\Rightarrow A = -\frac{V_m}{Z} \sin(\omega t_0 + \theta - \alpha)$$

$$i(t) = \underbrace{-\frac{V_m}{Z} \sin(\omega t_0 + \theta - \alpha) e^{-R/L(t-t_0)}}_{\text{S.R.}} + \underbrace{\frac{V_m}{Z} \sin(\omega t + \theta - \alpha)}_{\text{T.R.}}$$

condition for transient free response for case-(iii) is

$$V(t) = V_m \sin(\omega t + \theta), \quad t = t_0$$

$$\omega t_0 + \theta - \alpha = 0$$

$$\Rightarrow \omega t_0 = \alpha - \theta$$

$$\Rightarrow \omega t_0 = \tan^{-1}\left(\frac{\omega L}{R}\right) - \theta$$

case-(iv) :-

$$V(t) = V_m \cos(\omega t + \theta), \quad t = t_0$$

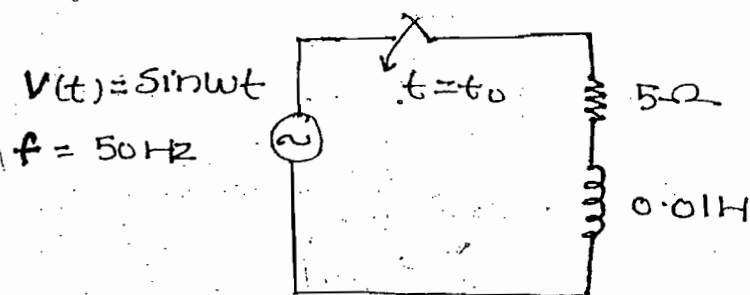
$$\omega t_0 = \tan^{-1}\left(\frac{\omega L}{R}\right) - \theta + \frac{\pi}{2}$$

	R-L	R-C	
(i) $V(t) = V_m \sin(\omega t + \theta)$ $t = 0$	$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$	$\theta = \tan^{-1}(\omega C)$	$\theta = \tan^{-1}(\omega R)$
(ii) $V(t) = V_m \cos(\omega t + \theta)$ $t = 0$	$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right) + \frac{\pi}{2}$	$\theta = \tan^{-1}(\omega C) + \frac{\pi}{2}$	$\theta = \tan^{-1}(\omega R) + \frac{\pi}{2}$
(iii) $V(t) = V_m \sin(\omega t + \theta)$ $t = t_0$	$\omega t_0 = \tan^{-1}\left(\frac{\omega L}{R}\right) - \theta$	$\omega t_0 = \tan^{-1}(\omega C) - \theta$	$\omega t_0 = \tan^{-1}(\omega R) - \theta$
(iv) $V(t) = V_m \cos(\omega t + \theta)$ $t = t_0$	$\omega t_0 = \tan^{-1}\left(\frac{\omega L}{R}\right) - \theta$	$\omega t_0 = \tan^{-1}(\omega C) - \theta + \frac{\pi}{2}$	$\omega t_0 = \tan^{-1}(\omega R) - \theta + \frac{\pi}{2}$

→ The above condition for series circuit for transient free response and also for parallel circuit.

→ In the RLC circuit it is not possible to obtain transient free response since circuit is having two energy storage element (simultaneously both charging and discharging action are present)

ques:- Find t_0 to obtain transient free response



Soln:-

$$\omega t_0 = \tan^{-1}\left(\frac{\omega L}{R}\right) - \theta$$

$$t_0 = \frac{\tan^{-1}\left(\frac{\omega L}{R}\right)}{\omega} \rightarrow \text{rad} \rightarrow \text{rad/s}$$

$$t_0 = \frac{32.1 \times \pi / 180}{100\pi} = 1.78 \text{ ms, Ans}$$

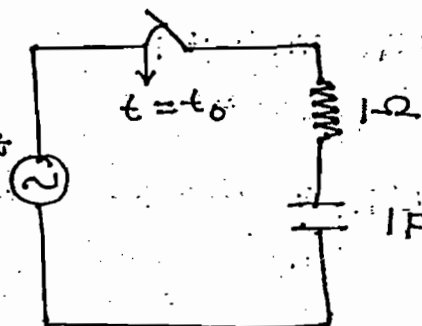
Ques:- Find t_0 to obtain transient free response.

Soln:- $\omega t_0 = \tan^{-1}(\omega RC) - \theta + \frac{\pi}{2}$

$$\Rightarrow t_0 = \frac{\tan^{-1}(1) + \frac{\pi}{2}}{1}$$

$$\Rightarrow t_0 = \frac{\frac{\pi}{4} + \frac{\pi}{2}}{1} = \frac{3\pi}{4} \quad (\pi = \frac{22}{7})$$

$$\Rightarrow t_0 = 7.35 \text{ s} \quad \text{Ans}$$



Laplace :-

$$1 \rightarrow \frac{1}{s}$$

$$e^{at} \rightarrow \frac{1}{s-a}$$

$$\sin \omega t \rightarrow \frac{\omega}{s^2 + \omega^2}$$

$$\cos \omega t \rightarrow \frac{s}{s^2 + \omega^2}$$

$$e^{at} \sin \omega t \rightarrow \frac{\omega}{(s-a)^2 + \omega^2}$$

$$e^{-at} \sin \omega t \rightarrow \frac{\omega}{(s+a)^2 + \omega^2}$$

$$e^{-at} \cos \omega t \rightarrow \frac{s+a}{(s+a)^2 + \omega^2}$$

$$\frac{df}{dt} \Rightarrow sF(s) - f(0)$$

$$u(t) \rightarrow \frac{1}{s}$$

$$\delta(t) \rightarrow 1$$

$$f(0^+) = \lim_{s \rightarrow \infty} sF(s) = \lim_{t \rightarrow 0} f(t)$$

→ Initial value theorem

Ques:- Find initial value of the following functions:-

$$F(s) = \frac{(s+1)(s+3)}{s(s+2)(s+4)}$$

Soln:- $f(0^+) = \lim_{s \rightarrow \infty} sF(s)$

$$= \lim_{s \rightarrow \infty} \frac{s \cdot s^2 \left(1 + \frac{1}{s}\right) \left(3 + \frac{3}{s}\right)}{s \cdot s^2 \left(2 + \frac{2}{s}\right) \left(4 + \frac{4}{s}\right)} = \frac{1}{2}$$

Ques:- Find initial value of the following function

$$f(t) = 3 + 8(t)$$

Soln:- $F(s) = \frac{3}{s} + 1$

Note:-

For the above function initial value theorem can't be applied since to apply the initial value theorem denominator power should be greater than Numerator power

Final Value Theorem:-

$$f(\infty) = \lim_{s \rightarrow 0} sF(s) = \lim_{t \rightarrow \infty} f(t)$$

Ques:- Find final value of the following function

(i) $F(s) = \frac{(s+1)(s+3)}{s(s+2)(s+4)}$

(ii) $f(t) = 3 + e^{2t}$

Soln:- (i) $f(\infty) = \lim_{s \rightarrow 0} sF(s)$

$$= \lim_{s \rightarrow 0} \frac{s(s+1)(s+3)}{s(s+2)(s+4)} = \frac{3}{8}$$

(ii) $f(t) = 3 + e^{2t}$

$$F(s) = \frac{3}{s} + \frac{1}{s-2}$$

$$V_C(t) = \frac{1}{C} \int_{-\infty}^t i dt$$

$$\Rightarrow V_C(t) = \frac{1}{C} \int_{-\infty}^{0^-} i dt + \frac{1}{C} \int_{0^-}^t i dt$$

$$\Rightarrow V_C(t) = V_C(0^-) + \frac{1}{C} \int_{0^-}^t i dt$$

At $t=0^+$

$$V_C(0^+) = V_C(0^-) + \frac{1}{C} \int_{0^-}^{0^+} i dt$$

$\underbrace{\int_{0^-}^{0^+} i dt}_{=0}$

$$V_C(0^+) = V_C(0^-) \quad \text{--- (I)}$$

$$V_C(0^+) = V_C(0^-) + \frac{1}{C} \int_{0^-}^{0^+} i dt$$

$$V_C(0^+) = V_C(0^-) + \frac{1}{C} \int_{0^-}^{0^+} \delta(t) dt$$

$\rightarrow \delta(t)$

Area = 1

$$\boxed{V_C(0^+) = \frac{1}{C}} \quad \text{--- (II)}$$

$$W_C(0^+) = \frac{1}{2} C V_C^2(0^+) = \frac{1}{2} C \left(\frac{1}{C}\right)^2$$

$$\boxed{W_C(0^+) = \frac{1}{2C}} \quad \text{--- (III)}$$

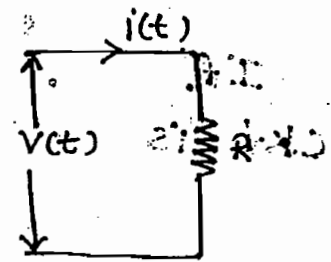
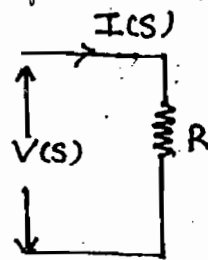
Note:-

→ From eq-(I) it is concluded that capacitor does not allow instantaneous changes for given i/p.

→ From eq-(II) & (III) it is concluded that capacitor allows instantaneous changes for current impulse function.

$$V(t) = i(t) R$$

$$V(s) = I(s) R$$

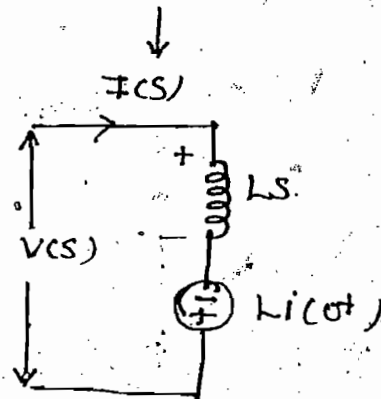
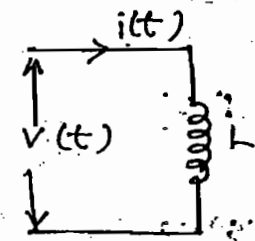


$$V = L \frac{di}{dt}$$

$$s = j\omega$$

$$V(s) = L [sI(s) - i(0^+)]$$

$$V(s) = L s I(s) - L i(0^+) \rightarrow \text{KVL}$$

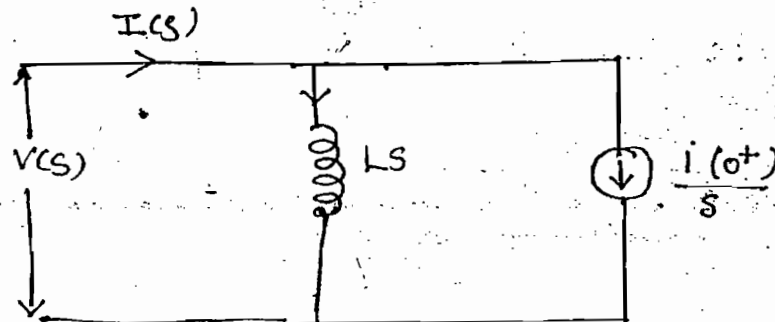


$$Z(s) = \frac{V(s)}{I(s)}$$

$$\Rightarrow \boxed{Z(s) = LS - \frac{L i(0^+)}{I(s)}}$$

$$L s I(s) = V(s) + L i(0^+)$$

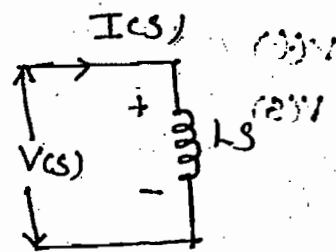
$$I(s) = \frac{V(s)}{Ls} + \frac{L i(0^+)}{Ls} \rightarrow \text{KCL}$$



$$Y(s) = \frac{I(s)}{V(s)}$$

$$\boxed{Y(s) = \frac{1}{Ls} + \frac{i(0^+)}{sV(s)}}$$

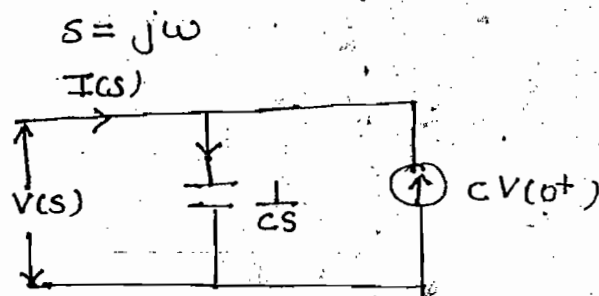
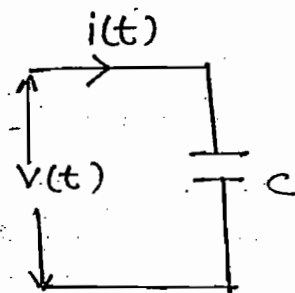
If $i(0^+) = 0$ then eq.
ckt is $\rightarrow$



$$i = C \frac{dV}{dt}$$

$$I(s) = C [sV(s) - V(0^+)]$$

$$\Rightarrow I(s) = \frac{V(s)}{1/Cs} - C V(0^+) \rightarrow \text{KCL}$$

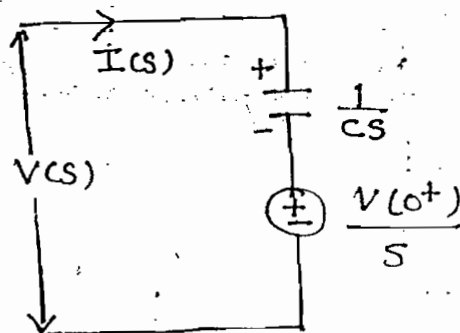


$$Y(s) = \frac{I(s)}{V(s)}$$

$$Y(s) = Cs - \frac{C V(0^+)}{V(s)}$$

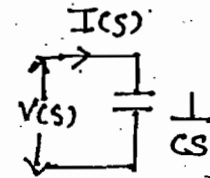
$$\frac{V(s)}{1/Cs} = I(s) + C V(0^+)$$

$$\Rightarrow V(s) = \frac{1}{Cs} I(s) + \frac{C V(0^+)}{Cs} \rightarrow \text{KVL}$$

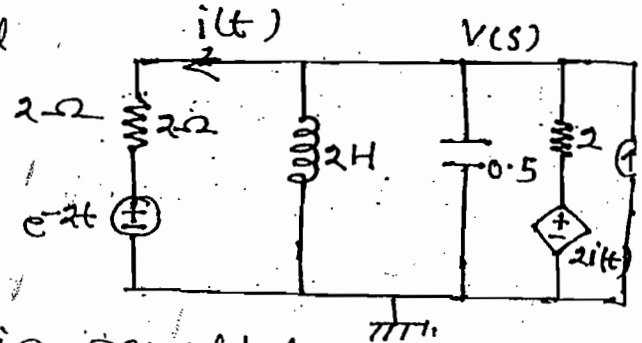


$$Z(s) = \frac{V(s)}{I(s)} = \frac{1}{Cs} + \frac{V(0^+)}{sI(s)}$$

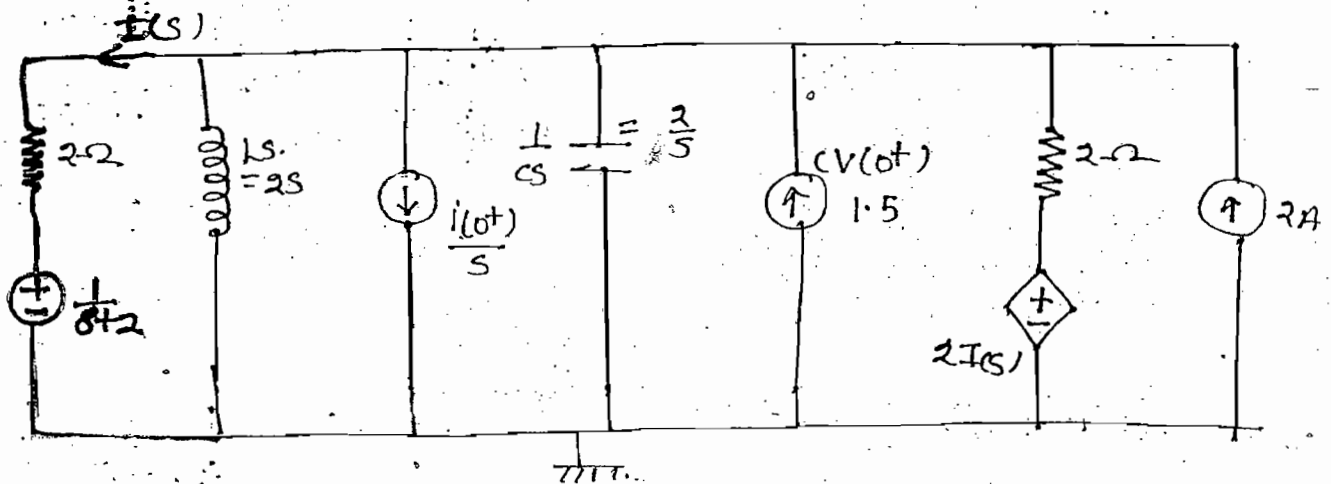
$$V(0^+) = 0 \rightarrow$$



Ques:- Find $V(s)$ when initial current of the inductor is $2A$ and initial voltage of the capacitor is $3V$



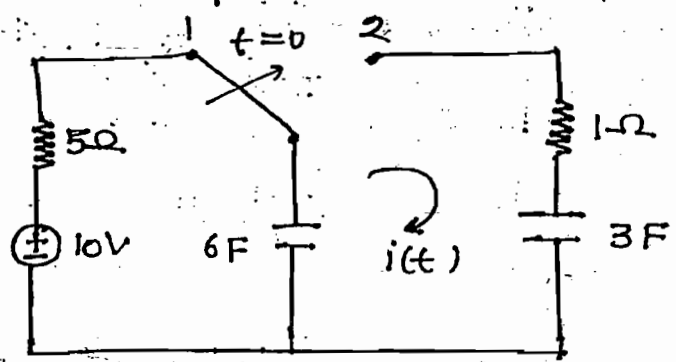
Soln:- When elements are in parallel then for simple calculation convert elements into current sources



$$\frac{V(s) - \frac{1}{s+2}}{2} + \frac{V(s)}{2s} + \frac{3}{s} + \frac{V(s)}{\frac{1}{Cs}} + \frac{V(s) - 2I(s)}{2} = 1.5 + 2 \quad \text{---(1)}$$

$$I(s) = \frac{V(s) - \frac{1}{s+2}}{2}$$

Ques:- Find $i(t)$ for $t > 0$



Soln:-

At $t = 0^-$

$$V_{C_6}(0^-) = 10V$$

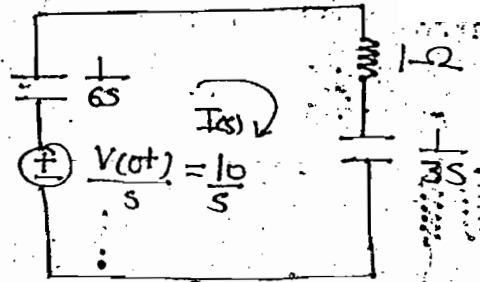
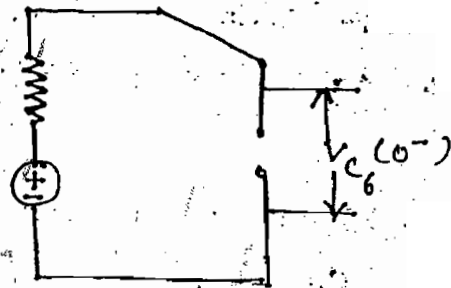
$$V_{C_3}(0^-) = 0V$$

At $t > 0$

$$I(s) = \frac{10/s}{\frac{1}{6s} + \frac{1}{3s} + 1}$$

$$i(t) = \mathcal{L}^{-1} I(s)$$

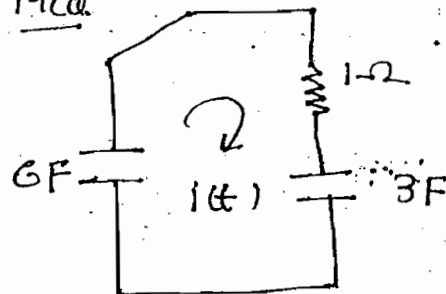
$$\Rightarrow i(t) = 10 e^{-t/2}$$



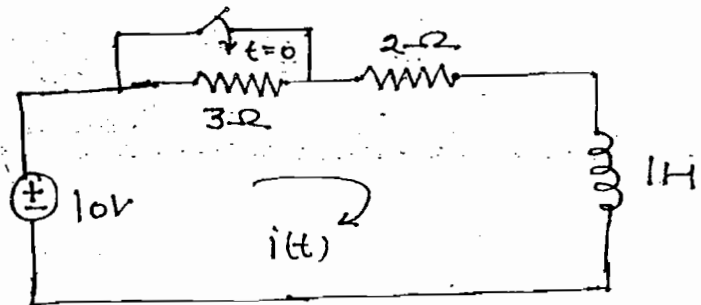
Verification:- $\rightarrow$ Method for MCA

$$i(t) = \frac{V_0}{R} e^{-t/RC}$$

$$i(t) = \frac{10}{1} e^{-t/2}$$



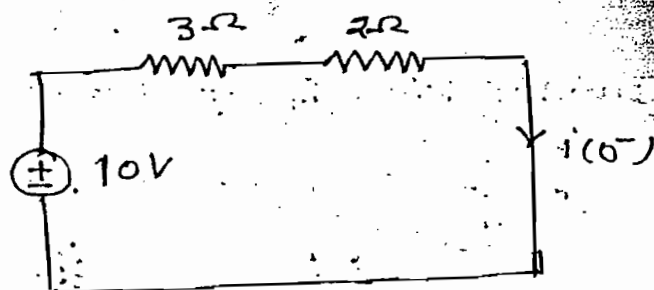
Ques:- Find $i(t)$ for $t > 0$



Soln:-

At $t = 0^-$

$$i(0^-) = \frac{10}{3+2} = 2A$$

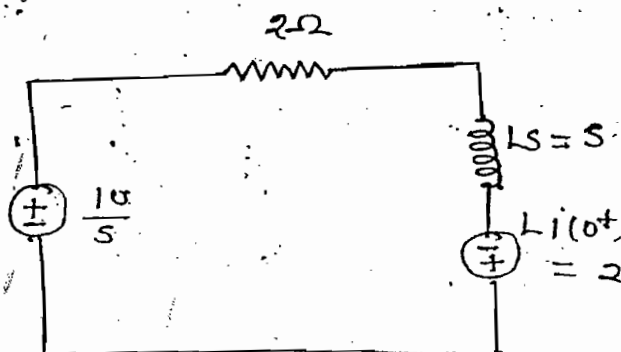


For $t > 0$

$$I(s) = \frac{\frac{10}{s} + 2}{2+s}$$

$$i(t) = \mathcal{L}^{-1} I(s)$$

$$i(t) = 5 - 3e^{-2t}$$

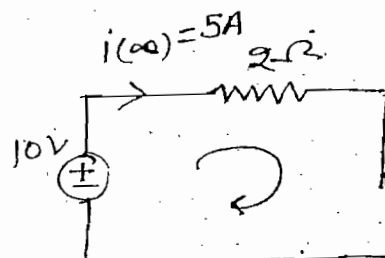


Alternate way:-

$$i(t) = [i(0^+) - i(\infty)]e^{-\frac{R}{L}t} + i(\infty)$$

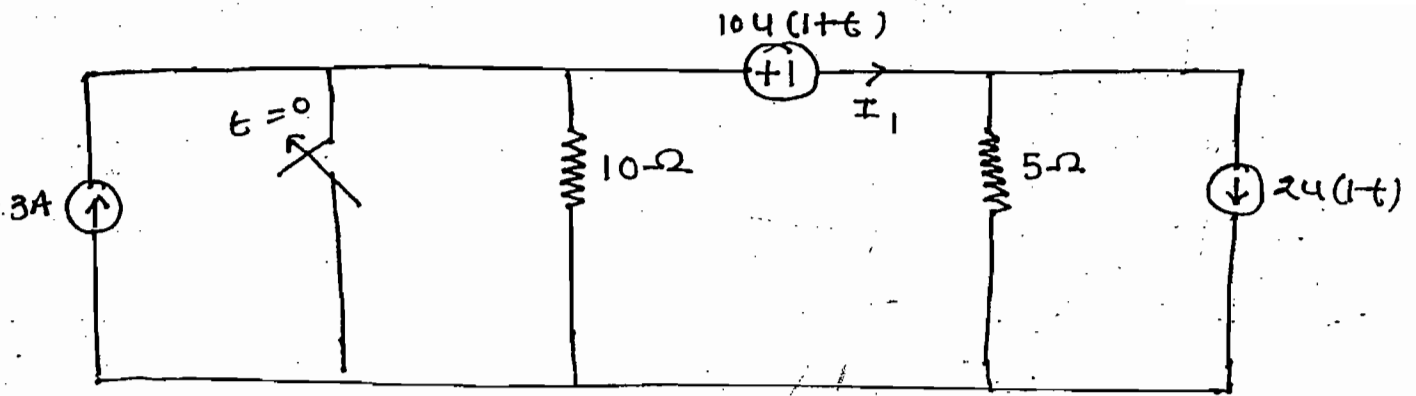
$$i(0^+) = i(0^-) = 2$$

$$i(t) = [2 - 5]e^{-2t} + 5$$



Lecture -11

Ques:- Find I_1 at $t = -2$ seconds:



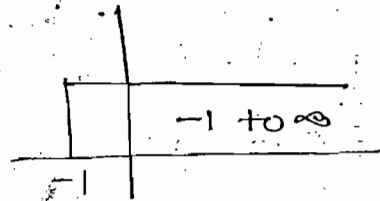
Soln:-

$$DC \rightarrow -\infty \text{ to } \infty$$

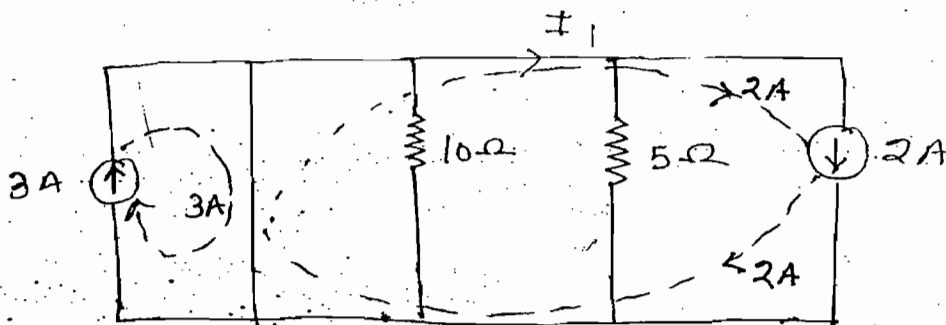
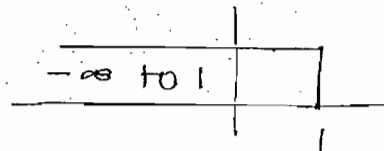
$$u(t) \rightarrow 0 \text{ to } \infty$$

$$u(-t) \rightarrow -\infty \text{ to } 0$$

$$u(1+t)$$

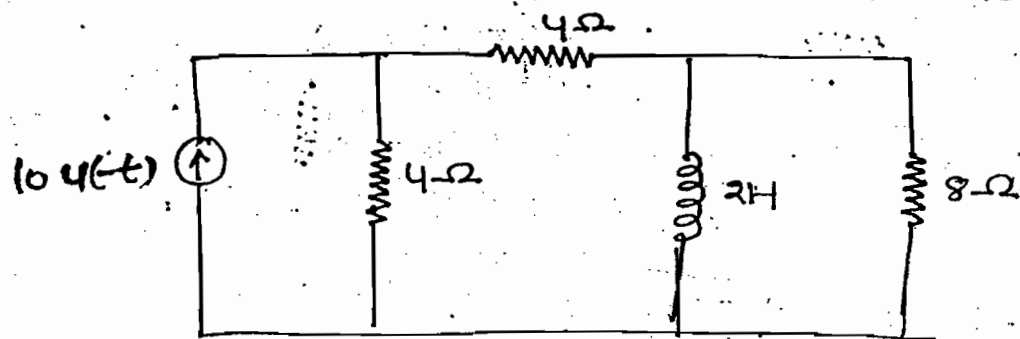


$$u(1-t)$$



$$I_1 = 2A$$

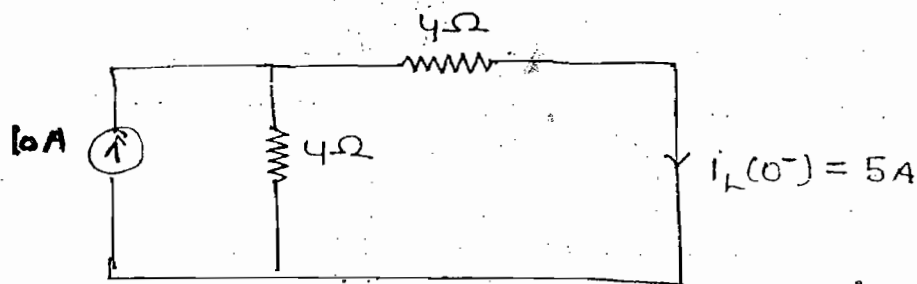
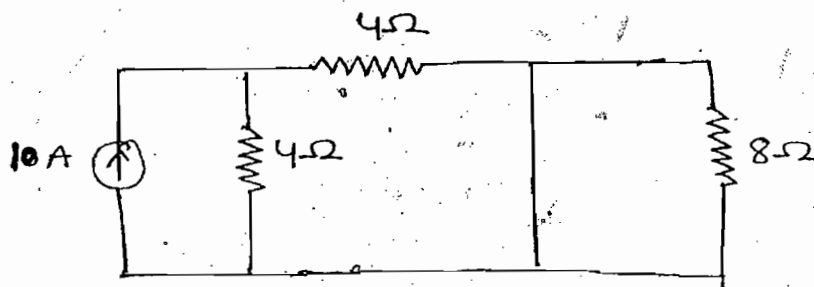
Ques:- Find current response in the inductor for $t > 0$



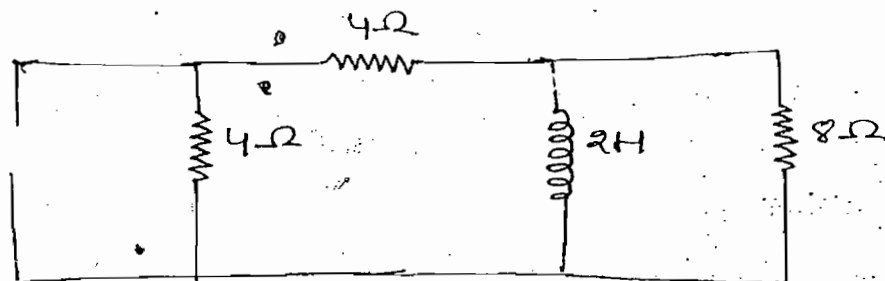
Soln:-

$t = 0^-$

$i(t) \rightarrow -\infty$ to 0



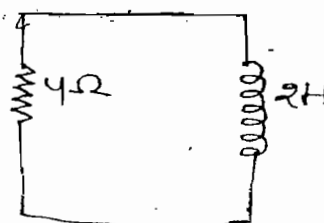
$t > 0$



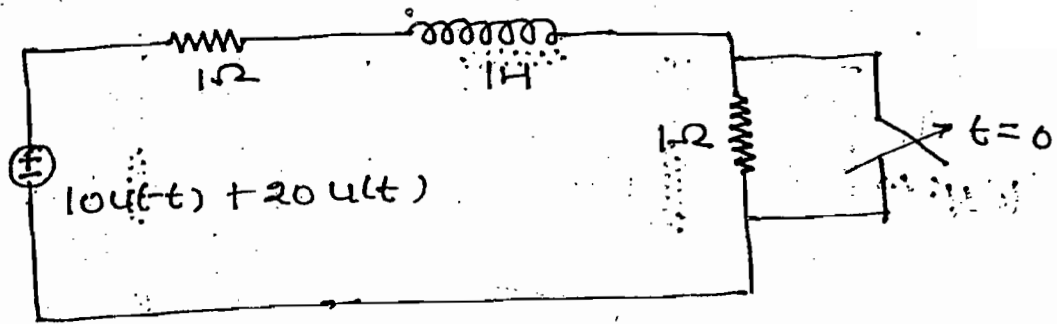
$$i(t) = I_0 e^{-R_L t}$$

$$i(t) = 5 e^{-2t}, \text{ Ans}$$

$$I_0 = i_L(0) = 5A$$



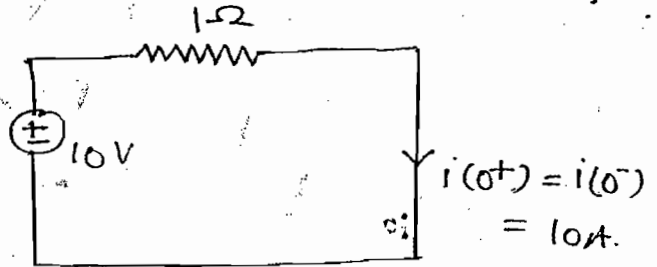
ques:- Find current in the inductor for $t > 0$



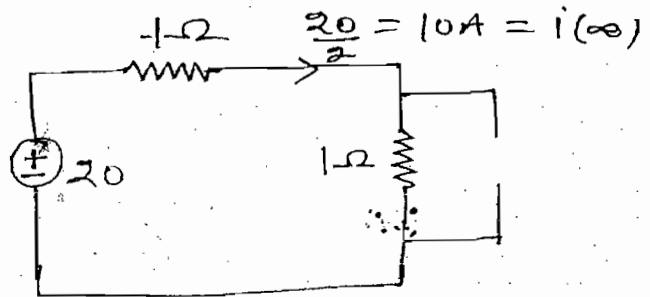
soln:- $i(t) = [i(0^+) - i(\infty)] e^{-\frac{R}{L}t} + i(\infty)$

$t = 0^-$

$$i(0^+) = i(0^-) = 10A$$



$t = \infty$



$$i(t) = [10 - 10] e^{-\frac{R}{L}t} + 10 = 10 \text{ Ans.}$$